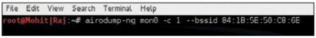

Figure 5: Command to set mon0 to Channel 1

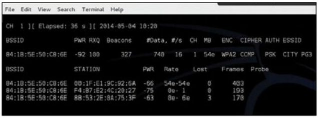

Figure 6: Showing all wireless devices connected to the AP

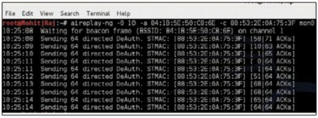

Figure 7: Sending the deauth packet

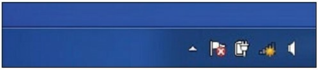
Figure 8: Wi-Fi signal gone

flag. The second deauthentication flag frame is sent from the AP to the victim. A local packet capture session is initiated using Wireshark to capture the frames generated by the attacker.

Who is behind the attack?

This attack is made at the data-link layer, which is associated with the MAC address. The book, 'Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet' (Second Edition) by Eoghan Casey, states that the data-link layer addresses (MAC addresses) are more easily identifiable than network layer addresses (e.g., IP addresses). This is because a MAC address is usually directly associated with the network interface card in a computer, whereas an IP address can be easily reassigned to different computers. However, in Wireshark-captured data, the source is the victim and the destination is the AP, and vice versa. Therefore, it is impossible to find out the attacker's identity.

So how do we detect the attack?

The deauthentication frame is sent by a station to another station when it wishes to terminate communications. When we manually disconnect from the AP, we can see three dauth packet after restarting AP three times as shown in figure 10. By using aireplay we have sent one deauth packet but on Wireshark, we captured 256 frames.

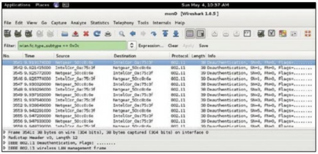
Figure 9: The packet flow

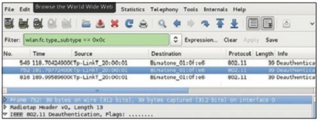

Figure 10: Deauth frame

Wireshark captured frames from one side and we have sent packets from the other side; so I can say that, from our side, 256/2 = 128 frames were sent. In this way, this attack also falls in the category of a DOS attack.

After seeing a large number of frames, a wireless intrusion detection system (WIDS) can raise the alarm.

At the user level, there is still no fool-proof way to prevent this attack. But at the organisation level, a WIPS/WIDS system like AirMagnet Enterprise can specifically detect these attacks, preventing major enterprise-wide damage. Going ahead, wireless cards or APs should have some mechanism to protect users from deauthentication attacks. END

References

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[2] Rupinder Cheema, Divya Bansal, Dr Sanjeev Sofat, June 2011. 'Deauthentication/Disassociation Attacks: Implementation and Security in Wireless Mesh Networks', International Journal of Computer Applications (0975 – 8887) Volume 23– No. 7

[3] Thuc D Nguyen, Duc H M Nguyen. August 3 -7, 2008, ' A light weight solution for defending against deauthentication /disassociation attacks on 802.11 networks', the 17th International Conference on Computer Communications and Networks, at St Thomas, US Virgin Islands, USA.

[4] *http://www.aircrack-ng.org/doku.php?id=aireplay-ng*

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